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Research Article

DIGITAL PROTOCOL IN THE AGE OF ARTIFICIAL INTELLIGENCE: TRANSFORMATION, CHALLENGES AND STRATEGIC RECOMMENDATIONS FOR PUBLIC SECTOR MANAGEMENT

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Abstract: The accelerating digitalisation of inter-state communication has produced a new field of institutional practice — digital protocol — which regulates the conduct of official interactions across electronic channels, virtual platforms and social media. Grounded in the network society theory and the author's published monographs on state protocol and the digital era, this paper offers both a conceptual examination and a multi-source empirical analysis of digital protocol as a strategic management challenge for the public sector. Drawing on North Macedonia's National Cybersecurity Strategy 2025-2028, the ECHO H2020 project, the ITU Guide to Developing a National Cybersecurity Strategy, the GFCE/Cybil Portal, the GCSCC Oxford Cybersecurity Capacity Maturity Model, and the MoI Digital Forensics Laboratory OKTIV (ISO/IEC 17025 | ENFSI | IACIS), the paper identifies five operational domains of digital protocol and derives seven evidence-based strategic recommendations for public sector managers. The central argument is that digital protocol is the functional transformation of traditional protocol — not its replacement — and requires institutional governance equivalent to that of its physical counterpart.

Keywords: digital protocol, digitalisation, public sector management, artificial intelligence, cybersecurity

1. INTRODUCTION

The digitalisation of state and diplomatic practice is no longer a future prospect — it is an operational reality. Video-conferencing platforms have replaced face-to-face summits, social media accounts have become official diplomatic channels, and artificial intelligence is beginning to support protocol planning, translation and event management. Yet the academic and institutional frameworks that govern protocol — the system of rules, hierarchies and symbolic practices through which states and public institutions project authority, manage relations and communicate intent — have adapted to this transformation with considerable delay (Bjola & Holmes, 2015; Smiljanov, 2018).

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Castells (2009) argued that in the network society, the most fundamental form of power lies in the ability to shape the human mind through communication networks. Digital protocol is precisely the institutionalised system through which public sector actors attempt to manage, control and project meaning across those networks in official contexts. It sits at the intersection of strategic management, public sector governance, diplomacy and information technology — thematic priorities of IMCSM26.

The paper makes three contributions. First, it provides an original conceptualisation of digital protocol as a strategic management domain grounded in network society theory. Second, it offers a multi-source case analysis — the ECHO H2020 project, the MoI Digital Forensics Laboratory (OKTIV), and North Macedonia's National Cybersecurity Strategy 2025-2028 — as concrete illustrations of institutional engagement with digital protocol. Third, it derives seven strategic recommendations for public sector managers.

2. THEORETICAL FRAMEWORK

Castells' theory of the network society (Castells, 2009, 2012) provides the foundational theoretical lens. Castells argues that power is exercised through the programming and switching of networks: those who control the protocols of communication networks hold structural power. This insight is directly applicable to digital protocol: the institution that masters the norms, formats and symbolic grammar of digital interaction gains a structural advantage in inter-institutional and inter-state relations.

Taddeo (2017) extends this analysis into the cybersecurity domain, arguing that digital conflicts are characterised by their informational nature: the target is not physical infrastructure but the information systems through which authority is projected. Digital protocol violations — hacked social media accounts, intercepted communications or AI-generated deepfakes — are therefore acts of informational warfare against the representational authority of public institutions. The International Telecommunication Union (ITU, 2025), as the United Nations specialised agency for digital technologies uniting 194 national governments, has tracked the growth of National Cybersecurity Strategies globally from 76 countries in 2018 to 136 today.

Bjola and Holmes (2015) provide the empirical bridge, demonstrating that the migration of diplomatic practice into digital space has not eliminated the need for protocol, but has transformed its form and created new vulnerabilities. The Global Forum on Cyber Expertise (GFCE), through the Cybil Portal (GFCE Cybil Portal, 2026), coordinates international cyber capacity building across over 260 members and partners. The Global Cyber Security Capacity Centre at Oxford (GCSCC, 2021) provides the Cybersecurity Capacity Maturity Model (CMM), which governments use to assess their digital governance maturity.

3. DIGITAL PROTOCOL: CONCEPTUAL FRAMEWORK AND FIVE DOMAINS

Protocol in its classical sense is a system of rules prescribing correct conduct in official and ceremonial contexts involving governments, nations and their representatives (Smiljanov, 2018). It is simultaneously a normative framework and a communicative instrument: it structures access, signals hierarchy, frames reciprocity and converts political intention into observable state behaviour. Smiljanov (2018) defines protocol as 'the face of the state' — emphasising its essential representational function.

Digital protocol extends this system into the electronic domain. Smiljanov (2020) identifies the fundamental continuity: each new communication technology — telephone, fax, email — required new protocol norms while preserving the underlying principles of hierarchy,

reciprocity, accuracy and respect. Digital protocol is therefore not a rupture with tradition but its functional adaptation.

Five intersecting domains constitute digital protocol: (1) digital communication etiquette — norms governing electronic correspondence, telephonic conduct and online conferencing; (2) virtual diplomatic encounters — adaptation of ceremonial functions to video-conferencing and hybrid formats; (3) social media as a protocol instrument — official platforms used for public diplomacy and political signalling; (4) AI in protocol management — emerging use of artificial intelligence for translation, scheduling and communication analysis; and (5) cybersecurity and protocol integrity — protection of official digital channels from interference, hacking and impersonation (Bjola & Holmes, 2015; ITU, 2025; Smiljanov, 2023).

4. KEY DIMENSIONS OF DIGITAL PROTOCOL

4.1. Digital communication etiquette

Electronic communication has become the primary medium of official correspondence. Smiljanov (2018) establishes clear protocol norms: official email must be treated as formal business correspondence; institutional email addresses are mandatory; the subject field must always be completed; and responses must be provided within 24 hours. The MoI Digital Forensics Laboratory OKTIV (Ministry of Interior of the Republic of North Macedonia — OKTIV, 2024) demonstrates that electronic communications leave permanent digital traces with legal weight — reinforcing the need for institutional discipline in digital communication.

Mobile phones and instant messaging applications present distinct challenges. Smiljanov (2018) notes that mobile phones must be silenced during official meetings and formal occasions. Conference calls require explicit consent before connecting additional participants. The emergence of WhatsApp, Telegram and Microsoft Teams for institutional communication has created unresolved protocol dilemmas that require clear institutional policy frameworks.

4.2. Virtual diplomatic encounters

The COVID-19 pandemic accelerated the normalisation of virtual diplomatic encounters. Heads of state, ministers and senior officials conducted summits, bilateral meetings and multilateral negotiations via Zoom, Microsoft Teams and Google Meet (Smiljanov, 2023). This transition raised fundamental protocol questions: Who controls the visual environment in a virtual encounter? How is precedence established in a grid view? How are ceremonial elements — national anthems, flag displays, formal welcomes — incorporated into digital formats?

Emerging institutional practice points to several principles. The host institution retains responsibility for the digital environment and must ensure technical quality, security and the preservation of formal elements. Dress code requirements remain in force. Background choices communicate institutional identity and must be managed with the same care as physical venue selection. Virtual diplomatic activities enable a more inclusive process but place new demands on protocol staff's technical competence (Smiljanov, 2023).

4.3. Social media as a protocol instrument

Social media has become an official diplomatic channel. Castells (2009, 2012) theorised this as 'mass self-communication' — a form of communication that is simultaneously mass

(global reach) and self-generated (bypassing institutional gatekeepers). The Canada-Saudi Arabia Twitter dispute of 2018 — in which a government tweet on human rights triggered a diplomatic crisis and the expulsion of the Canadian ambassador — illustrates how a social media protocol violation can produce consequences equivalent to a formal diplomatic incident (Smiljanov, 2023).

The Trump administration's use of Twitter as a primary foreign policy communication channel — bypassing formal diplomatic channels and protocol procedures — demonstrated both the power and the danger of unregulated social media conduct by senior officials (Smiljanov, 2023). Social media posts by officials constitute official communications subject to protocol norms and institutions must establish clear social media protocol policies specifying what may be published, by whom and under what circumstances.

4.4. Artificial intelligence in protocol management

Artificial intelligence is beginning to alter the operational landscape of protocol management. Smiljanov (2023) identifies emerging applications: AI assists in analysing large volumes of information, identifying trends and generating predictive analyses; machine learning provides decision-support information; and AI is automating administrative tasks in protocol offices, freeing staff for strategic work. Natural language processing enables real-time translation and cultural mediation — a significant advancement given that language and cultural misunderstanding represent a primary source of protocol errors.

However, AI integration introduces serious risks. The deepfake threat — generation of realistic audio or video of officials making statements they did not make — represents an existential challenge to the authenticity guarantees at the heart of diplomatic protocol. The 2022 deepfake of President Zelensky calling on soldiers to surrender during active conflict provides a case study of how AI-generated protocol violations can be weaponised (Taddeo, 2017). Institutions must develop governance frameworks, verification procedures and human oversight mechanisms.

5. CASE STUDIES: NORTH MACEDONIA IN ACTION

North Macedonia's national context for digital protocol development is framed by the National Cybersecurity Strategy 2025-2028, adopted by the Ministry of Digital Transformation (Ministry of Digital Transformation of the Republic of North Macedonia, 2024). Developed in alignment with the ITU Guide (ITU, 2025) — adopted by 136 countries globally — the Strategy articulates a vision of 'a secure, resilient and trustworthy digital environment' operationalised through an Action Plan covering five strategic pillars: capacity development, critical infrastructure protection, cybercrime, awareness and international cooperation.

A concrete case study of institutional engagement with digital protocol at the European level is provided by the ECHO project (H2020, Grant No. 830943), in which the Military Academy 'General Mihailo Apostolski' in Skopje participated as an official partner in a 30-member consortium from 16 European countries with a budget of €15,987,285 (ECHO Consortium, 2019-2023). ECHO developed the Early Warning System (E-EWS) for sharing cybersecurity incidents among trusted partners, the Federated Cyber Range (E-FCR) for training in realistic simulated environments, and the Information Sharing Model D3.3 (ECHO Consortium, 2022). The Macedonian Information Agency covered the author's related monograph 'The Digital Era' (Macedonian Information Agency, 2023), providing national media validation of the research.

The most operationally concrete case study is the Digital Forensics Laboratory of the Ministry of Interior — OKTIV (Ministry of Interior of the Republic of North Macedonia — OKTIV, 2024). This laboratory processes digital evidence from computers, mobile phones, tablets, GPS devices and drones. It is accredited to ISO/IEC 17025:2018, is a member of ENFSI (European Network of Forensic Science Institutes, 71 laboratories from 38 countries), participates in IACIS-certified proficiency testing, and cooperates with INTERPOL, EUROPOL, SEESAC and NABIS. Research by the Australian Strategic Policy Institute (ASPI, 2023) and the Norwegian Institute of International Affairs (NUPI, 2022) confirms that institutions with developed digital protocol frameworks are significantly more resilient to informational threats.

6. RISKS, VIOLATIONS AND SECURITY IMPLICATIONS

Digital protocol violations carry consequences equivalent to — and in some cases more serious than — their physical counterparts. Smiljanov (2023) identifies principal risk categories: diplomatic incidents arising from social media posts; security breaches through compromised official digital channels; reputational damage from protocol errors visible to global audiences; and erosion of institutional credibility through inconsistent digital conduct. The 2017 hacking of the US Department of Defense Twitter account and the wrong flag incident at the official reception of the Austrian Chancellor in Skopje in 2018 illustrate these dimensions (Smiljanov, 2023).

Taddeo (2017) provides the theoretical framework for understanding digital protocol violations as acts of informational conflict. Advanced Persistent Threats (APTs) targeting diplomatic missions, the 'Operation Aurora' attacks, and Russian interference in the 2016 US elections (Mueller, 2019) demonstrate the vulnerability of diplomatic networks to sophisticated intrusion. The human factor — phishing, weak passwords, unsecured channels — remains the primary vulnerability, confirmed by the ECHO project's cybersecurity training framework (ECHO Consortium, 2019-2023).

Disinformation and manipulation of official digital channels represent a third risk category. Castells (2009) theorised this as a form of counter-power in the network society: actors who cannot compete in conventional diplomatic channels use communication networks to reprogramme public perception. The FIRST (2024) standards provide technical protocols for inter-institutional incident communication — a direct application of digital protocol principles to cybersecurity events.

7. STRATEGIC RECOMMENDATIONS FOR PUBLIC SECTOR MANAGEMENT

Seven strategic recommendations can be derived for public sector managers. First, institutionalise digital protocol as a formal strategic management domain — not a technical matter delegated to IT departments, but a governance function with dedicated policy frameworks, trained personnel and regular review procedures equivalent to those governing physical protocol (Smiljanov, 2018). Second, invest in digital protocol literacy across all levels of institutional hierarchy, recognising that digital communications are governed by the same principles of accuracy, hierarchy and diplomatic consequence as formal written correspondence (Smiljanov, 2023).

Third, develop social media protocol policies specifying what may be published, who may publish in the institution's name, what review processes apply and how violations will be addressed. Fourth, establish secure digital communication infrastructure including encrypted

channels, multi-factor authentication and incident response procedures aligned with FIRST (2024) standards. Fifth, adapt ceremonial and representational functions to hybrid and virtual formats without abandoning their symbolic content — the ECHO Federated Cyber Range (ECHO Consortium, 2019-2023) provides a model for training in simulated digital protocol environments. Sixth, develop AI governance frameworks ensuring human oversight, deepfake verification procedures and ethical deployment guidelines, informed by ECHO and the GCSCC CMM (GCSCC, 2021).

Seventh, participate in international cooperation mechanisms for digital protocol standardisation. North Macedonia's NCS 2025-2028 (Ministry of Digital Transformation of the Republic of North Macedonia, 2024), developed in alignment with the ITU framework (ITU, 2025), provides the national anchor. The GFCE/Cybil ecosystem (GFCE Cybil Portal, 2026), ECHO network (ECHO Consortium, 2019-2023), FIRST (2024) and DiploFoundation (2024) together constitute the international support ecosystem. OKTIV (Ministry of Interior of the Republic of North Macedonia — OKTIV, 2024) provides the operational implementation infrastructure. The seven recommendations constitute a roadmap for integrating these frameworks into a coherent digital protocol governance system.

8. CONCLUSION

Digital protocol is the frontier on which the authority, credibility and security of public sector institutions are increasingly determined. Castells' insight (2009) — that power in the network society lies in the ability to shape meaning through communication networks — provides the theoretical foundation for understanding why digital protocol matters strategically: it is the institutionalised system through which public sector actors manage and project meaning in the networks that increasingly constitute the space of governance.

The empirical case studies demonstrate that this is not an abstract concern. North Macedonia's participation in the ECHO H2020 project, the operation of an ISO/IEC 17025-accredited Digital Forensics Laboratory within the MoI, the adoption of a National Cybersecurity Strategy aligned with the ITU framework, and the national media coverage of the author's monograph on the digital era collectively demonstrate that the institutional conditions for systematic digital protocol development are already in place.

The paper's central theoretical contribution is the conceptualisation of digital protocol as the functional transformation of traditional protocol, grounded in network society theory and illustrated through empirical case analysis. The five domains, three-tier risk framework and seven strategic recommendations constitute a comprehensive and actionable framework for public sector institutions. Digital protocol literacy must be treated as a core strategic management competency — not an optional enhancement — and this paper provides both the theoretical argument and the practical roadmap for making it so.

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